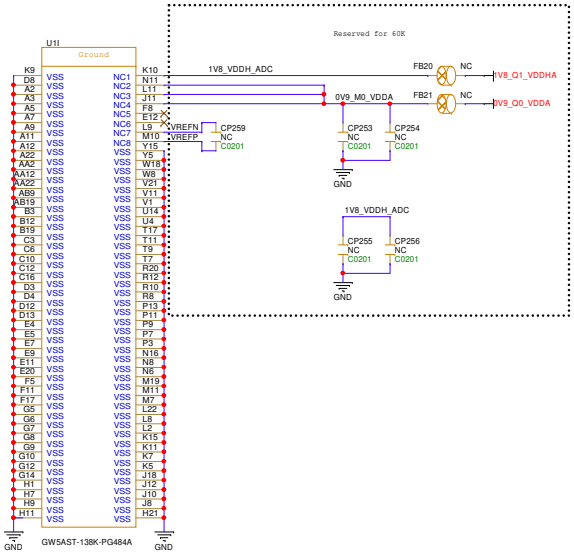
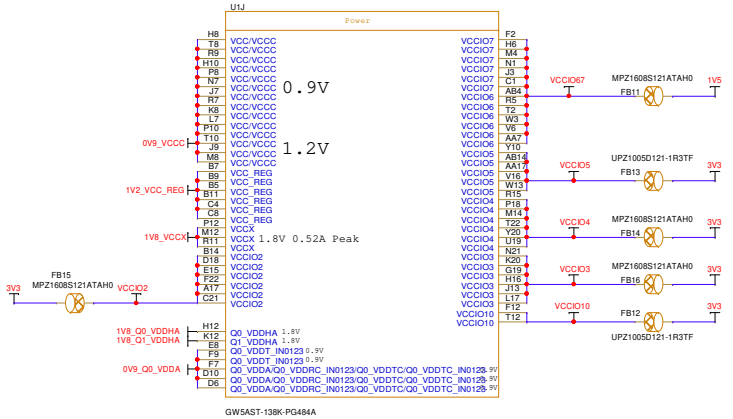
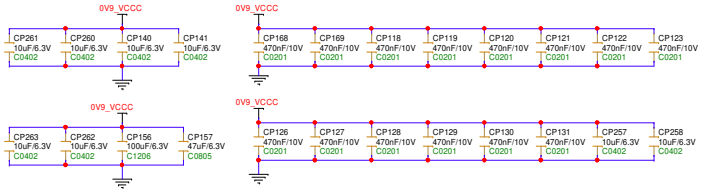
# FPGA POWER

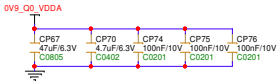


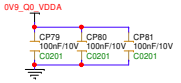



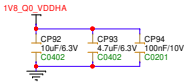



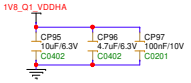


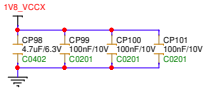


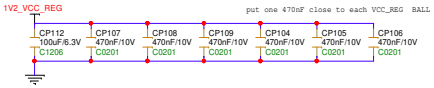


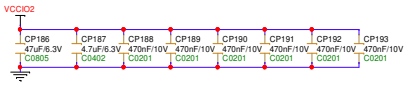



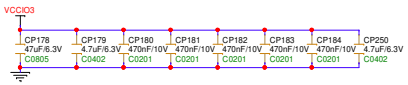



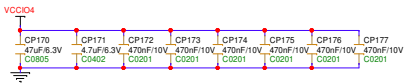



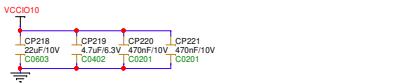



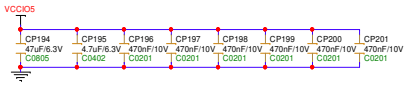



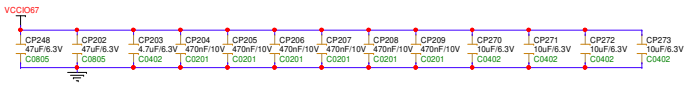



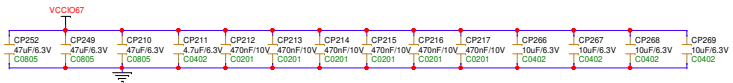



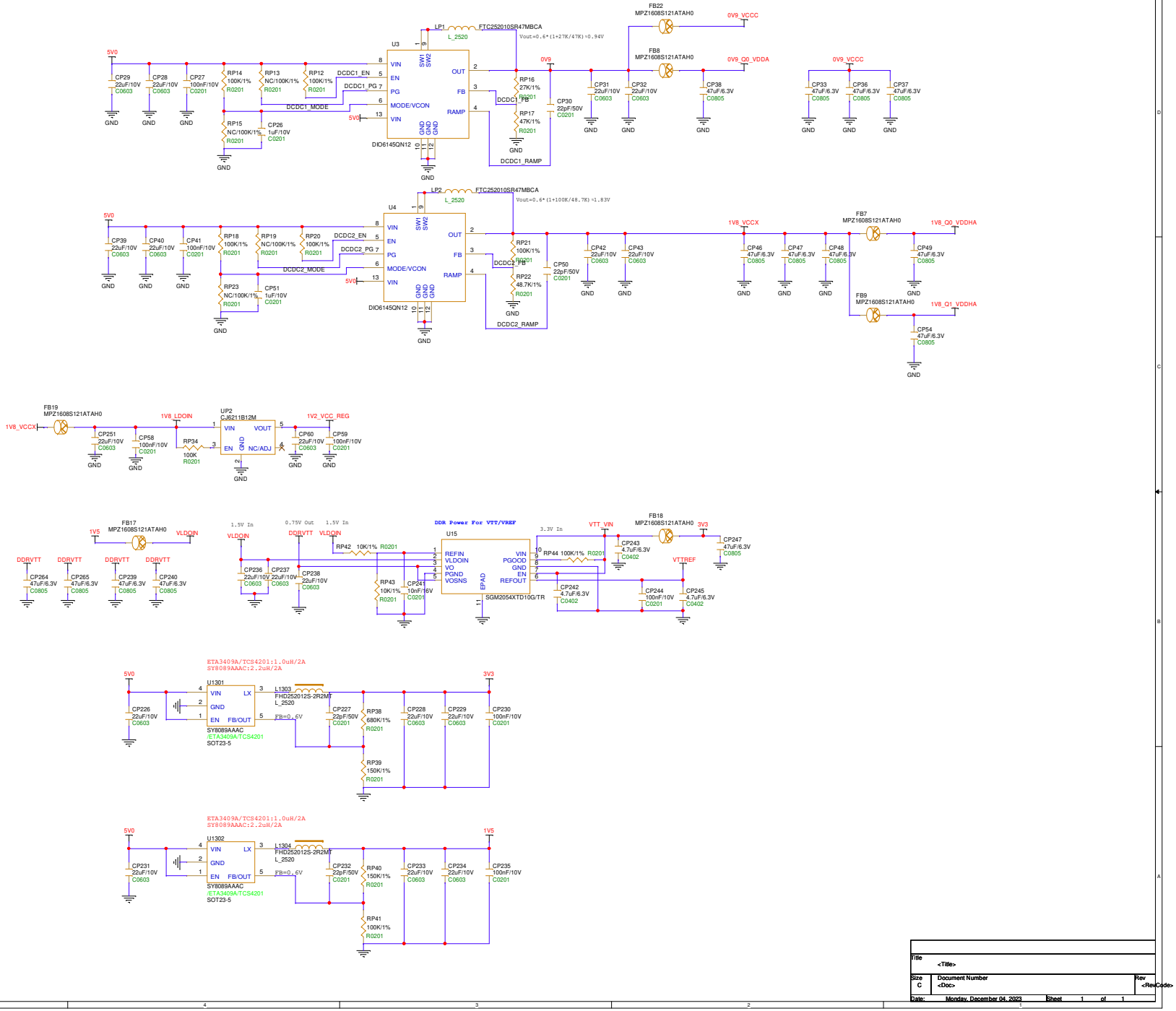



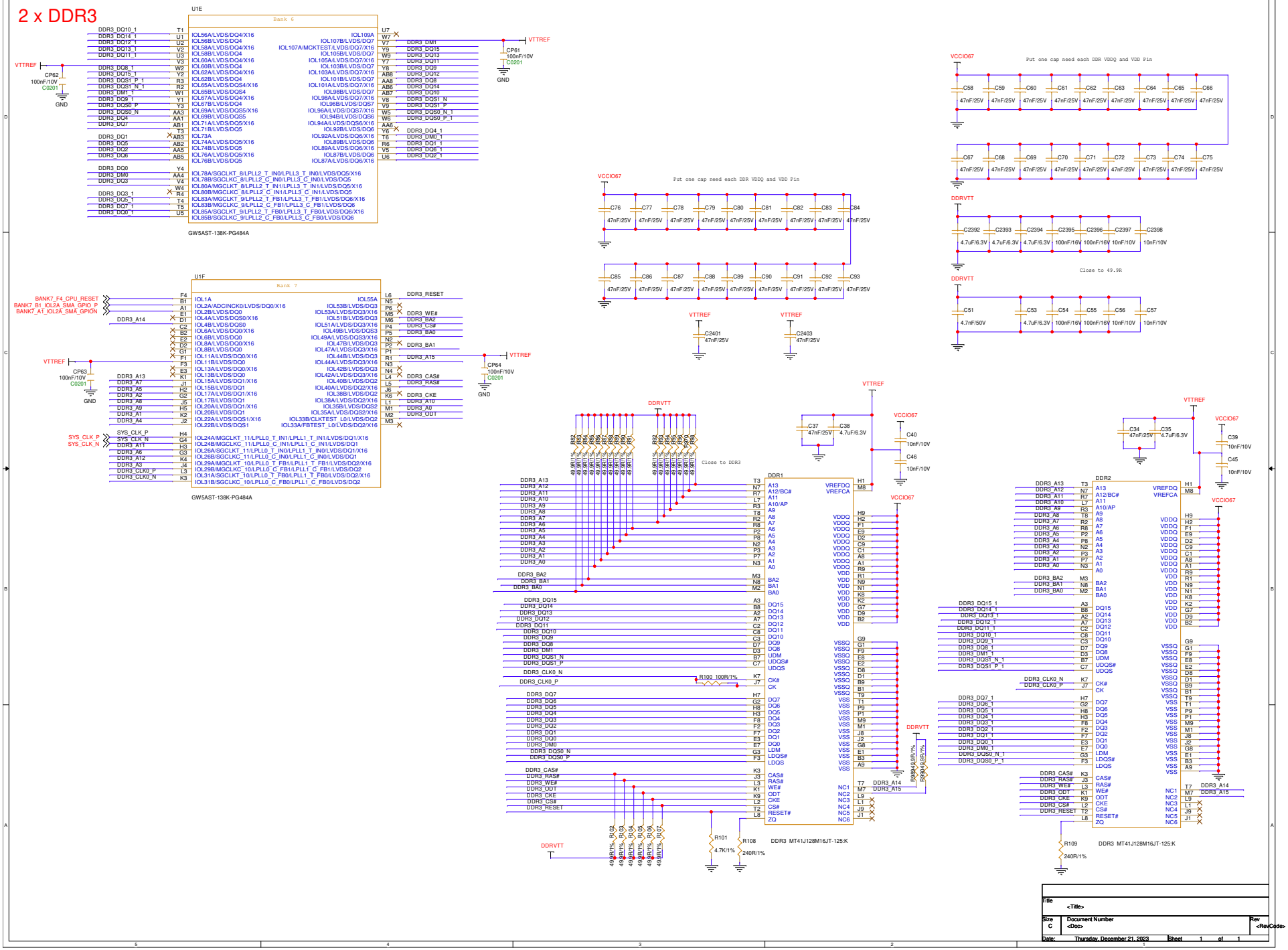

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|---|---|---|
| Title | <Title> | |
| Size: C | Document Number: <Doc> | Rev: <RevCode> |
| Date: Monday, December 04, 2023 | Sheet 1 of 1 | |

# Board Power

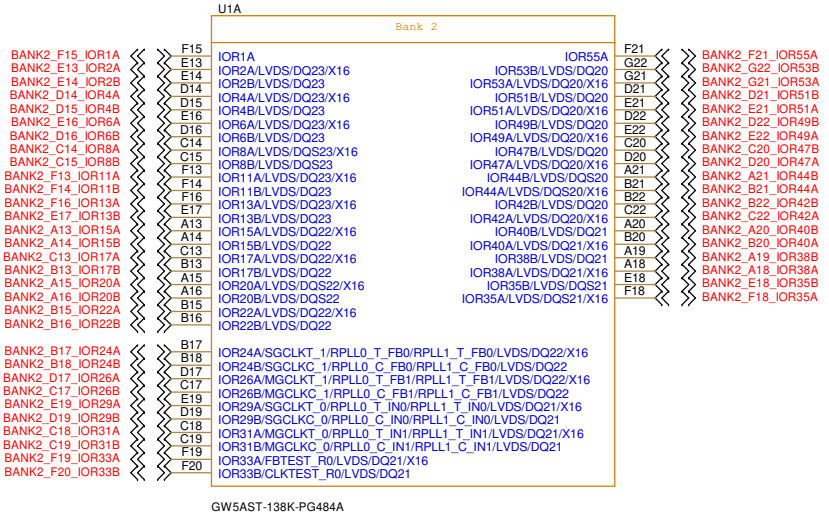



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| Title | <Title> | |
| Size C | Document Number <Doc> | Rev <RevCode> |
| Date: Monday, December 04, 2023 | Sheet 1 of 1 | |

# 2 x DDR3


U1E
Bank 6
GW5AST-138K-PG484A
U1F
Bank 7
GW5AST-138K-PG484A
DDR1
DDR2
DDR3_MT41J128M16JT-125:K
VTTREF
VCCIO67
DDRVTT
Put one cap need each DDR VDDQ and VDD Pin
Close to DDR3
Close to 49.9R
Title
<Title>
Size C
Document Number <Doc>
Rev <RevCode>
Date: Thursday, December 21, 2023
Sheet 1 of 1

# BANK2/3



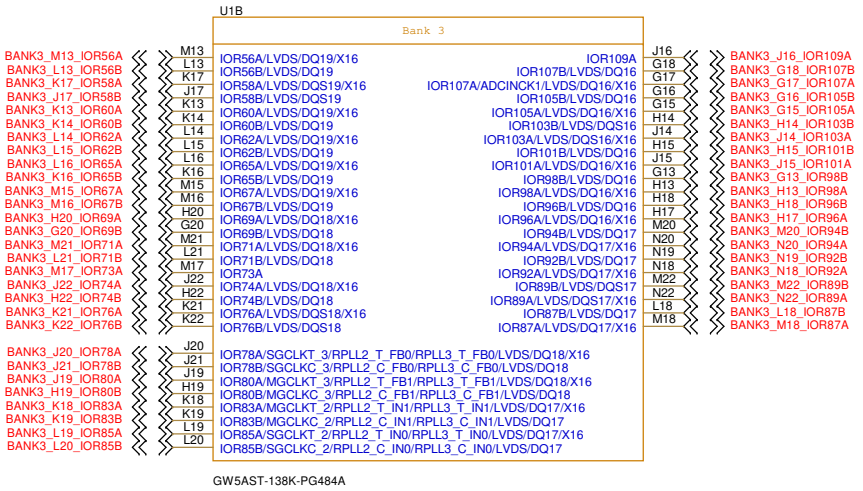
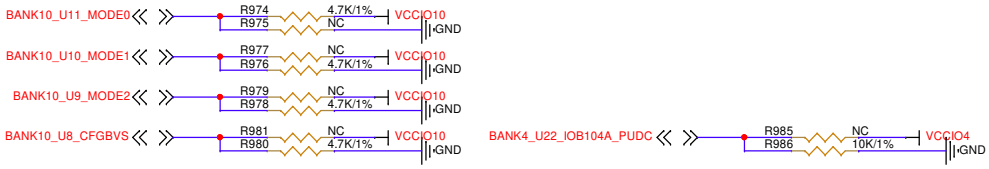




**表 4-2 GW5AT-138/GW5AST-138 配置模式选择**

| 配置模式 | MODE[2:0][1] | Bus Width | 相关说明 |
|---|---|---|---|
| JTAG | XXX/101[3] | - | 外部 Host 通过 JTAG 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置 |
| MSPI[2] | 001 | x1,x2,x4 | FPGA 作为 Master，通过 SPI 接口从外部 Flash（或其他器件）读取配置数据进行配置 |
| SSPI[2] | 010 | x1,x4 | 外部 Host 通过 SPI 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置 |
| Master SERIAL | 000 | x1 | FPGA 作为 Slave 以前，通过 DIN 接口从外部读取配置数据进行配置 |
| Slave SERIAL | 111 | x1 | 外部 Host 通过 DIN 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置 |
| Master CPU | 100 | x8,x16,x32 | FPGA 作为 Slave 以前，通过 DBUS 接口从外部读取配置数据进行配置 |
| Slave CPU | 110 | x8,x16,x32 | 外部 Host 通过 DBUS 接口对 GW5AT & GW5AST 系列 FPGA 产品进行配置 |

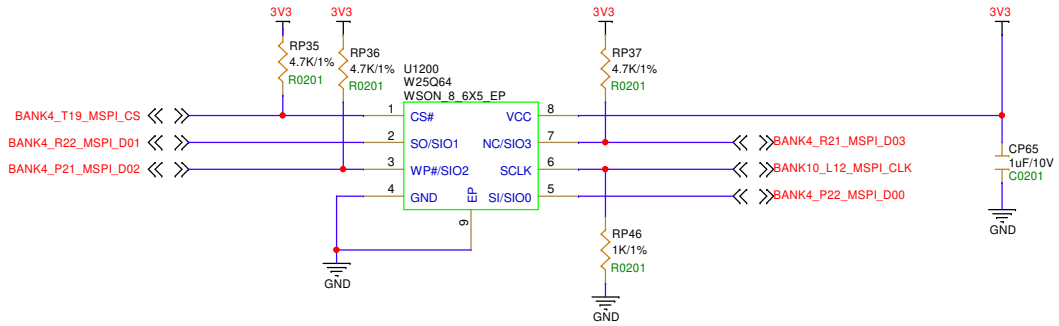
注！

- [1]对于一些 MODE 管脚没有全部封装出来的器件，需要查看 PINOUT 手册确认 MODE 脚状态。
- [2] SSPI 和 MSPI 模式的 SPI 接口是互相独立的。
- [3] JTAG 配置模式与 MODE[2:0]输入值无关；当 MODE 设置为 101 时，只有 JTAG 接口生效。

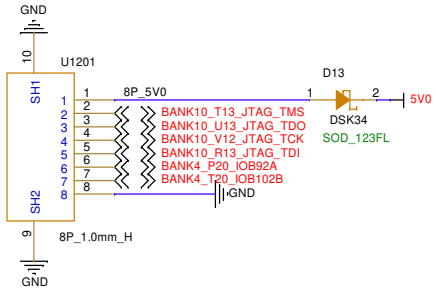



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| Title | <Title> | |
| Size B | Document Number <Doc> | Rev <RevCode> |
| Date: Friday, May 10, 2024 | Sheet 1 of 1 | |

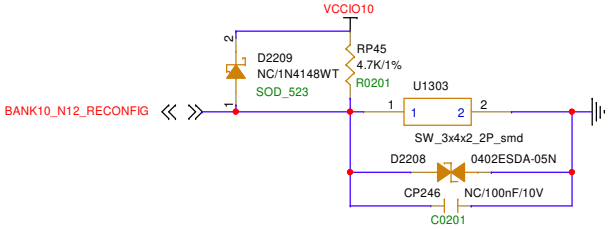
# MSPI FLASH

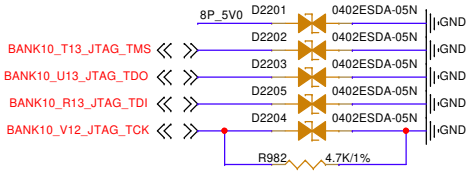


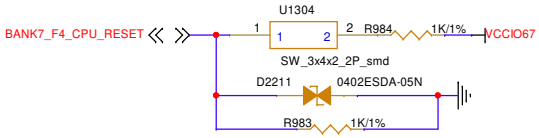

# JTAG Connector



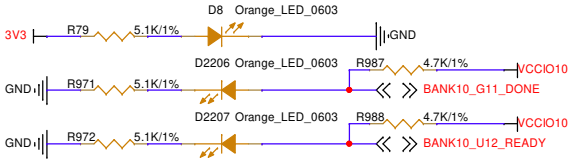

# Button



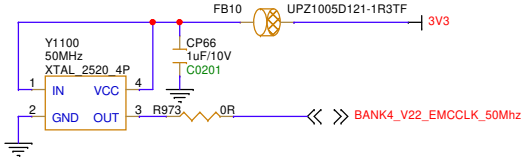




# LED

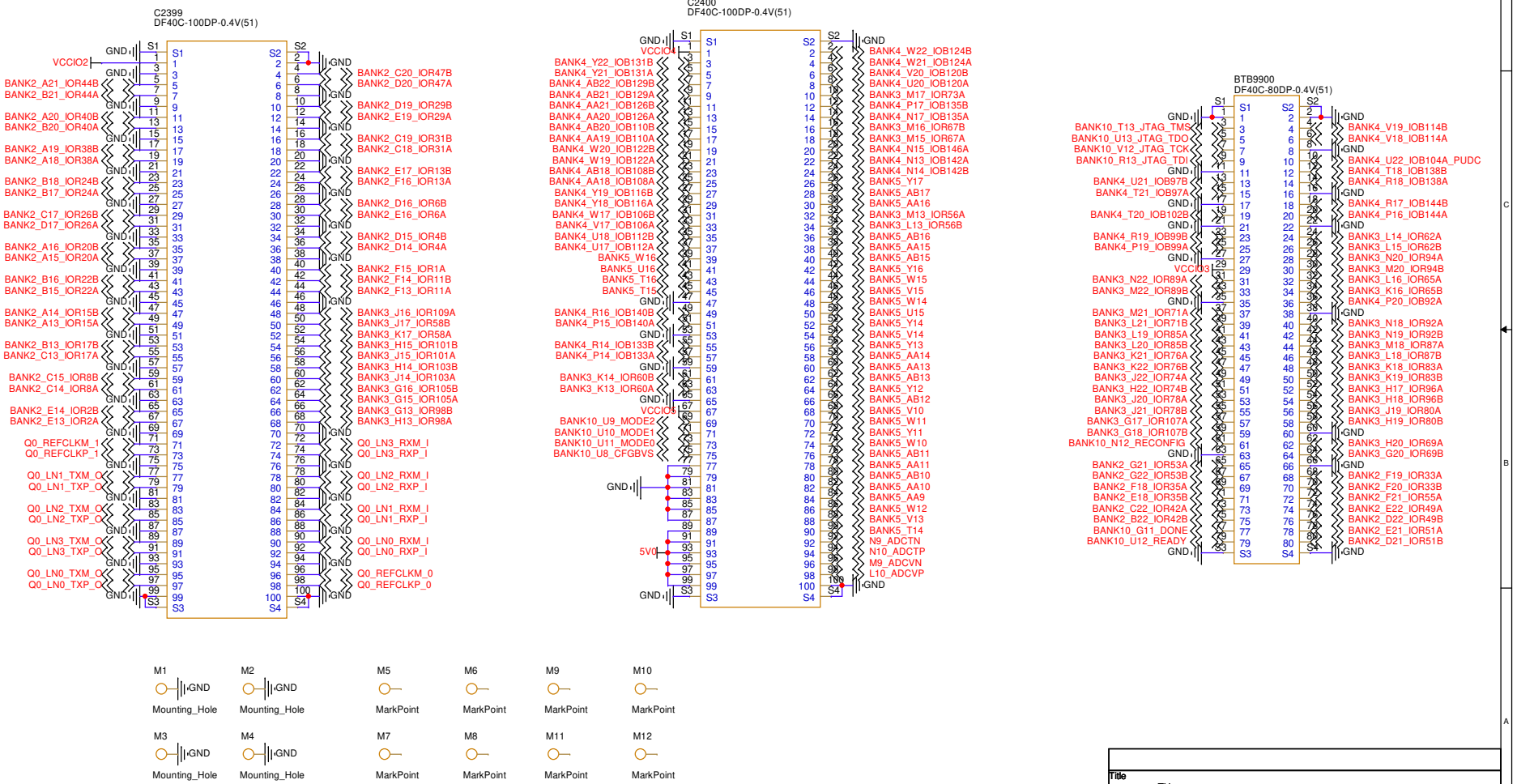



# Crystal oscillator




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| Title <Title> | | |
| Size B | Document Number <Doc> | Rev <RevCode> |
| Date: Friday, May 10, 2024 | Sheet 1 of 1 | |

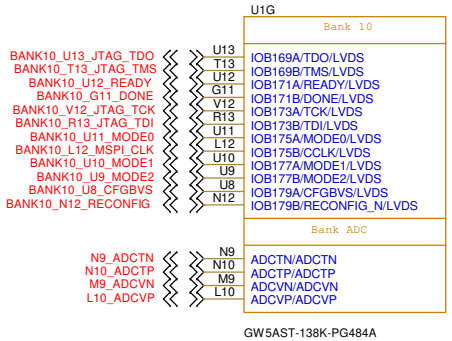
# BTB Connector

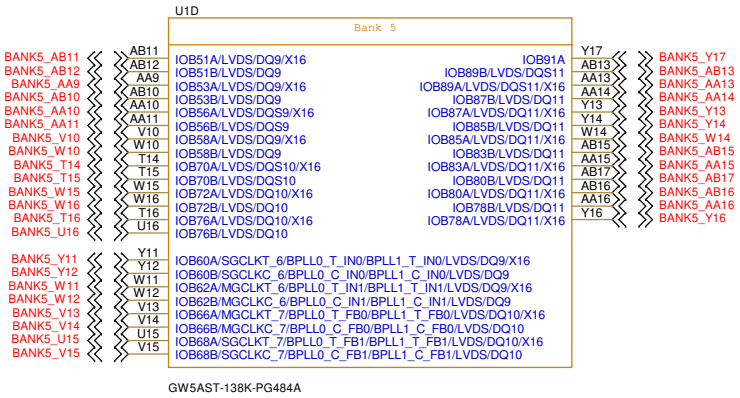


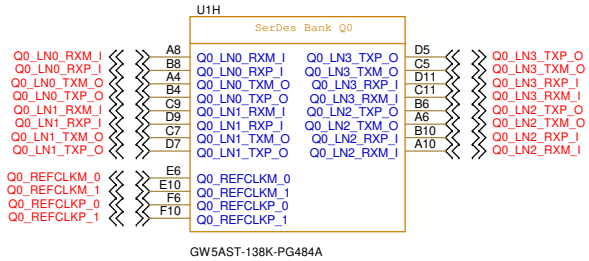

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| Title | <Title> | |
| Size B | Document Number <Doc> | Rev <RevCode> |
| Date: Thursday, December 07, 2023 | Sheet 1 of 1 | |

# BANK4/5/10/Q0/ADC




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|---|---|---|
| Title | <Title> | |
| Size B | Document Number <Doc> | Rev <RevCode> |
| Date: Wednesday, November 08, 2023 | Sheet 1 of 1 | |